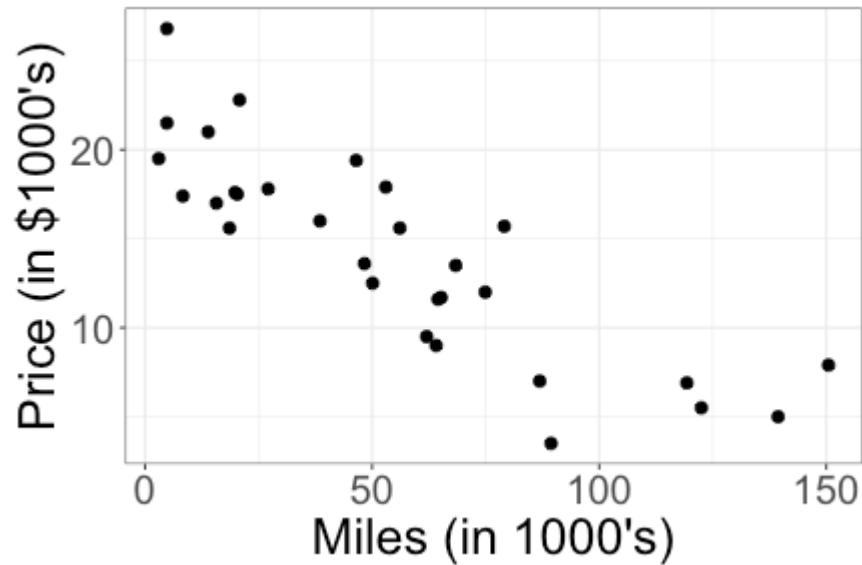


Inference for linear regression slope

Inference

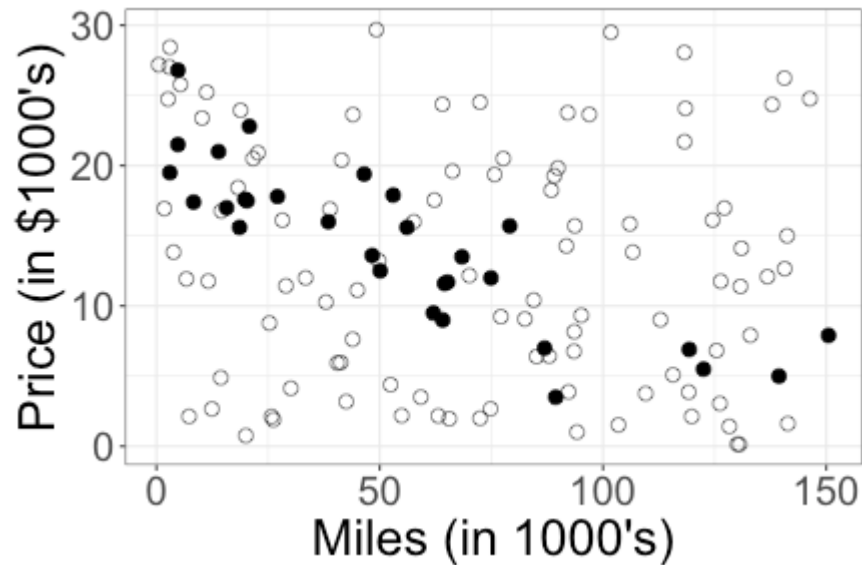
Is there a relationship between mileage and price for used Honda Accords?



It looks like it, but is that just a fluke?

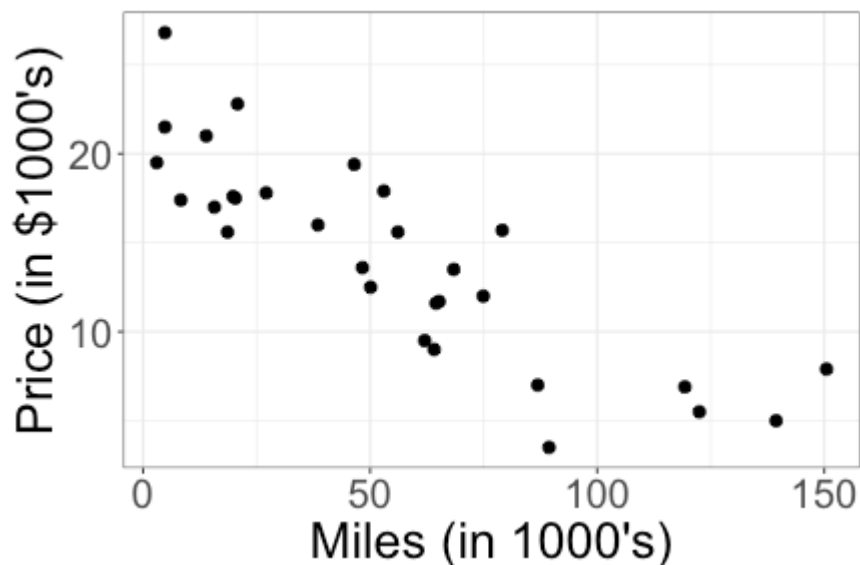
Is the observed relationship a fluke?

What if the population scatterplot looks like this?



Translating the question

Is there a relationship between mileage and price?

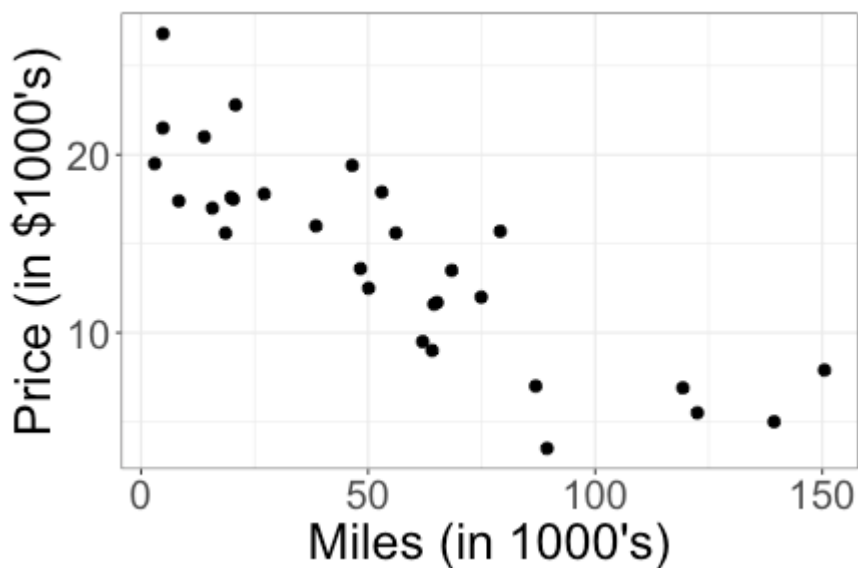


Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

If there is no relationship between mileage and price, what is true about my model?

Translating the question

Is there a relationship between mileage and price for used Honda Accords?

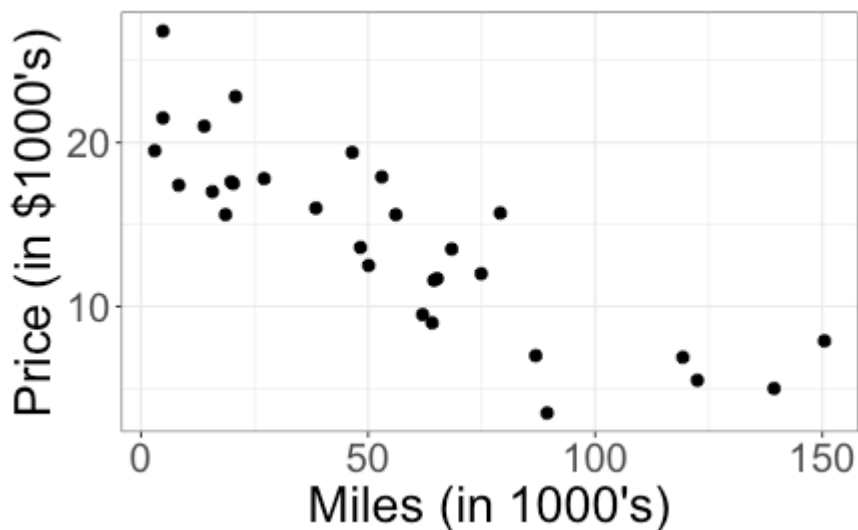


Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Question: Is $\beta_1 = 0$?

Hypothesis testing

$$\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon \quad \text{Is } \beta_1 = 0 ?$$



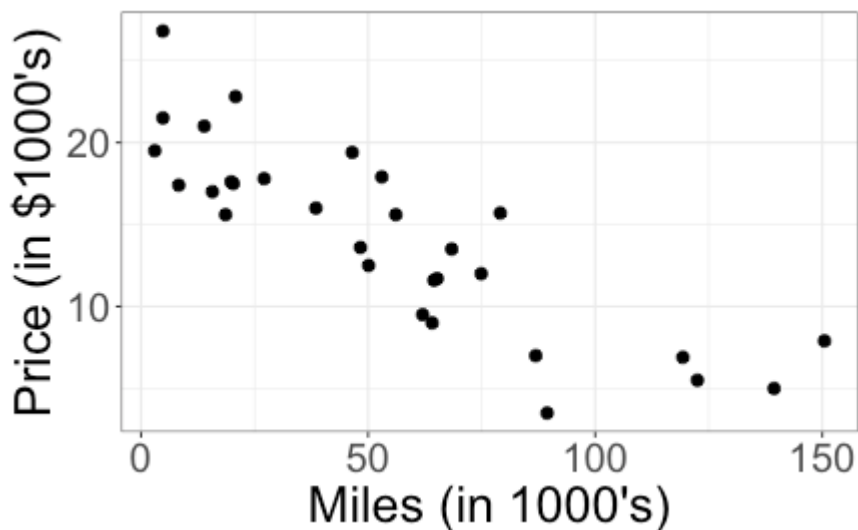
In statistics, we try to address these kinds of questions with *hypothesis testing*.

Hypothesis testing

What steps does a hypothesis test involve?

Hypothesis testing

$$\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon \quad \text{Is } \beta_1 = 0 ?$$



In statistics, we try to address these kinds of questions with *hypothesis testing*.

What do you think our null hypothesis (H_0) and alternative hypothesis (H_A) should be?

Specifying hypotheses

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Question: Is $\beta_1 = 0$?

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

- + The *null hypothesis*, H_0 , is that there is no relationship
- + The *alternative hypothesis*, H_A , is that there is a relationship
- + Our hypotheses are stated in terms of one or more model parameters

Specifying hypotheses

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

What if I want to know whether there is a *positive* relationship between mileage and price?

Specifying hypotheses

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

What if I want to know whether there is a *positive* relationship between mileage and price?

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 > 0$$

Estimating the relationship

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Estimated line: $\widehat{\text{price}} = 20.8 - 0.12 \text{ miles}$

If $\beta_1 = 0$, how unusual would $\hat{\beta}_1 = -0.12$ be?

Estimating the relationship

Model: $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Estimated line: $\widehat{\text{price}} = 20.8 - 0.12 \text{ miles}$

If $\beta_1 = 0$, how unusual would $\hat{\beta}_1 = -0.12$ be?

Need to know how $\hat{\beta}_1$ varies from sample to sample. How do we measure the variability of a statistic?

How unusual is the observed slope?

$$H_0 : \beta_1 = 0$$

Standardize: $t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}}$

where $SE_{\hat{\beta}_1}$ is the estimated standard error (standard deviation) of $\hat{\beta}_1$

How unusual is the observed slope?

$$H_0 : \beta_1 = 0$$

Standardize: $t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}}$

Accord data:

```
accord_lm <- lm(Price ~ Mileage, data = AccordPrice)
summary(accord_lm)
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096    0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198    0.0141   -8.50 3.06e-09 ***
```

How unusual is the observed slope?

$$H_0 : \beta_1 = 0$$

Standardize: $t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}}$

Accord data:

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096    0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198    0.0141   -8.50  3.06e-09 ***
```

What does a value of $t = -8.50$ mean?

How unusual is the observed slope?

Accord data:

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096      0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198      0.0141   -8.50 3.06e-09 ***
```

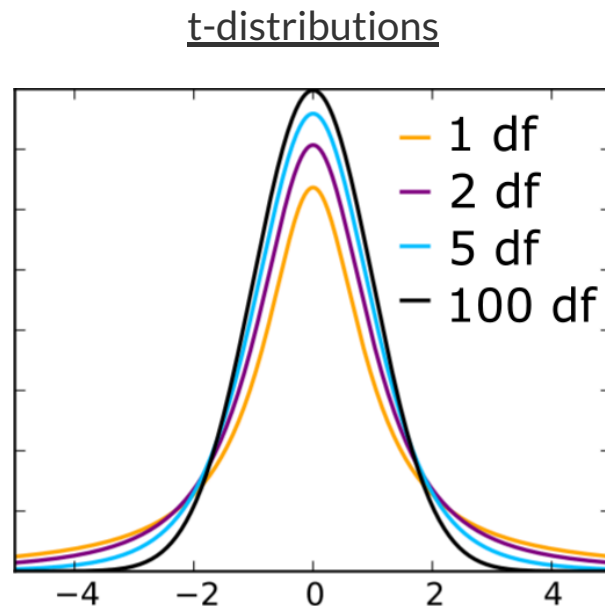
The observed value of $\hat{\beta}_1$ (-0.12) is 8.5 standard deviations below the value of β_1 hypothesized in $H_0 : \beta_1 = 0$

Is 8.5 standard deviations a lot?

How unusual is the observed slope?

$$t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}}$$

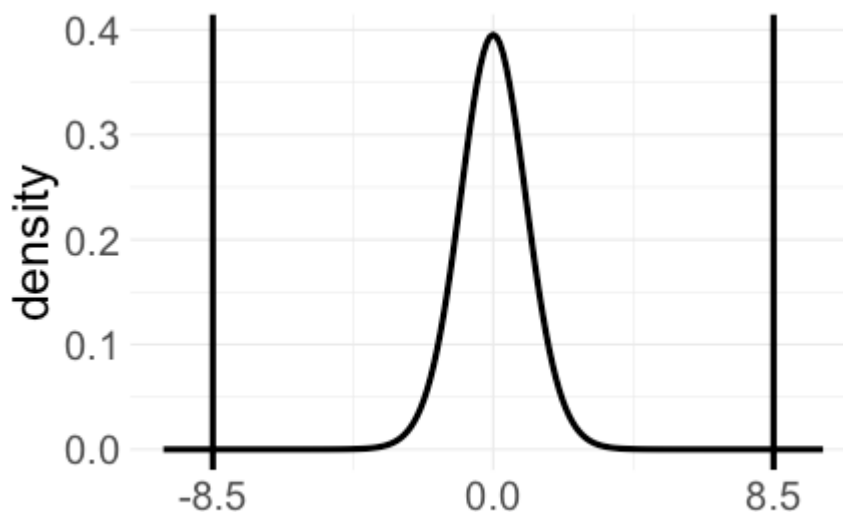
If $\beta_1 = 0$, then t follows a t -distribution with $n - 2$ degrees of freedom (df)



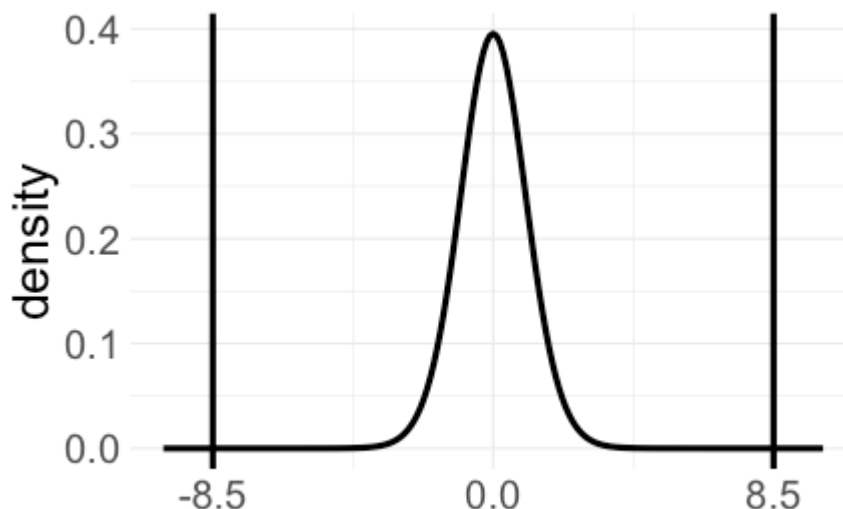
How unusual is the observed slope?

$$t = \frac{-0.12}{0.014} = -8.5, \quad n - 2 = 28$$

t_{28} distribution:



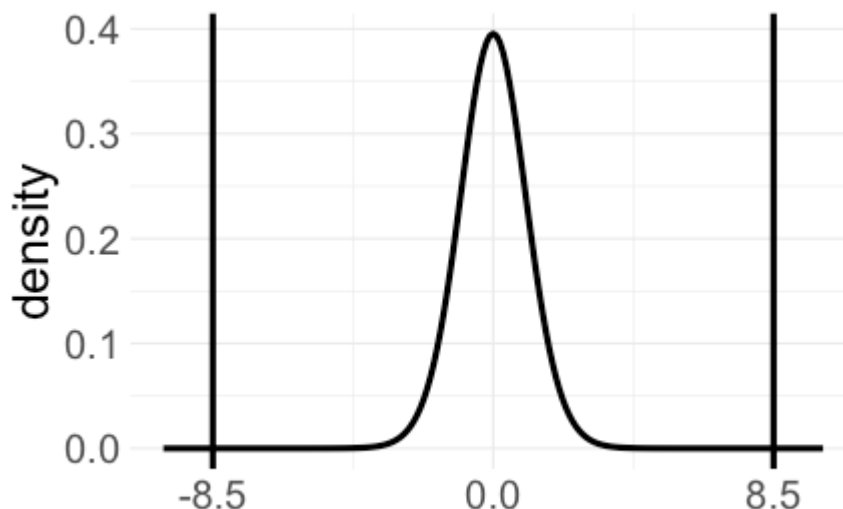
How unusual is the observed slope?



What fraction of the time do we see $t < -8.5$ or $t > 8.5$?

```
pt(-8.5, df=28, lower.tail = TRUE) +  
  pt(8.5, df=28, lower.tail = FALSE)
```

How unusual is the observed slope?

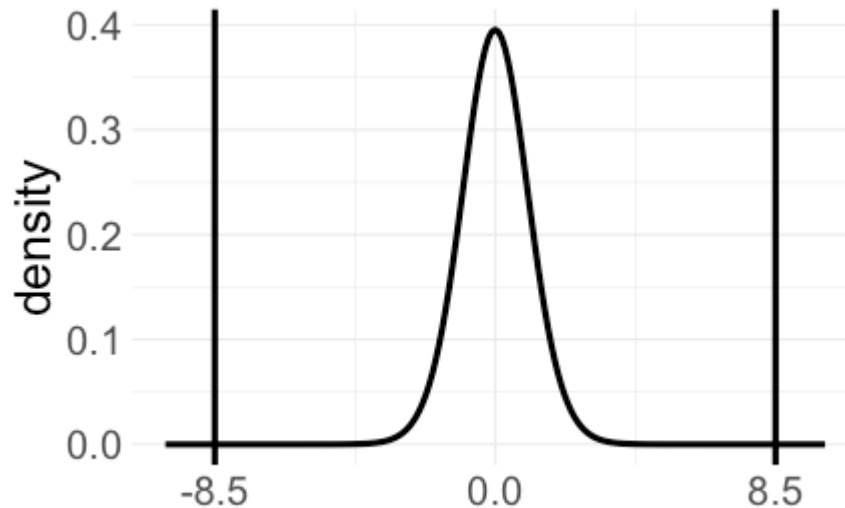


```
pt(-8.5, df=28, lower.tail = TRUE) +  
  pt(8.5, df=28, lower.tail = FALSE)
```

```
## [1] 3.056239e-09
```

If $\beta_1 = 0$, the probability of observing $t < -8.5$ or $t > 8.5$ is approximately 0.00000000003

How unusual is the observed slope?



- + If $\beta_1 = 0$, the probability of observing $t < -8.5$ or $t > 8.5$ is approximately 0.0000000003
- + $p\text{-value} = 0.0000000003$

Putting it together

Is there a relationship between mileage and price for used Honda Accords?

Model: $\text{price} = \beta_0 + \beta_1 \text{mileage} + \varepsilon$

Question: Is $\beta_1 = 0$?

Hypotheses:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Putting it together

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

Estimated line: $\widehat{\text{price}} = 20.8 - 0.12 \text{ mileage}$

Test statistic:
$$t = \frac{\hat{\beta}_1}{SE_{\hat{\beta}_1}} = \frac{-0.12}{0.014} = -8.5$$

p-value: 0.0000000003

Conclusion: The observed slope of -0.12 would be very unusual if there were no relationship between mileage and price for used Honda Accords, so we have strong evidence that there is a relationship between mileage and price.

In R

```
accord_lm <- lm(Price ~ Mileage, data = AccordPrice)
summary(accord_lm)
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  20.8096      0.9529   21.84  < 2e-16 ***
## Mileage      -0.1198      0.0141  -8.50 3.06e-09 ***
```

- + R automatically reports p-values for tests of the form

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

- + *Don't* rely on R output for other tests, such as

- + $H_0 : \beta_1 = 1 \quad H_A : \beta_1 \neq 1$

- + $H_0 : \beta_1 = 0 \quad H_A : \beta_1 > 0$

Class activity

Work on the activity (handout), then we will discuss as a class.

Class activity

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

Research question: is there a relationship between sparrow weight and wing length?

What are the null and alternative hypotheses?

Class activity

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

Coefficients:

	Estimate	Std. Error
(Intercept)	8.75465	1.39601
Weight	1.31341	0.09756

What is my test statistic for these hypotheses?

Class activity

$$\text{wing length} = \beta_0 + \beta_1 \text{weight} + \varepsilon$$

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

Coefficients:

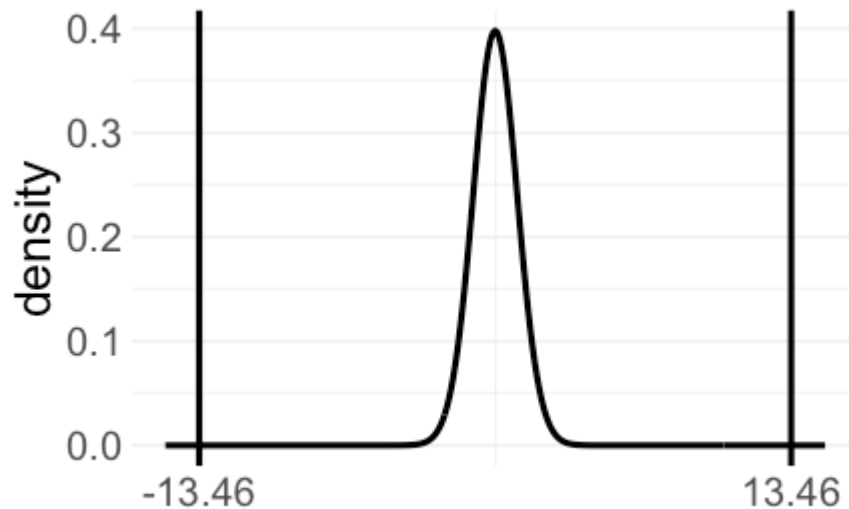
	Estimate	Std. Error
(Intercept)	8.75465	1.39601
Weight	1.31341	0.09756

$$t = \frac{\hat{\beta}_1 - 0}{SE_{\hat{\beta}_1}} = 13.46$$

Is this an "unusual" value of t if H_0 is true?

Class activity

$$t = 13.46 \quad n - 2 = 114$$



```
pt(-13.46, 114, lower.tail = TRUE) +  
  pt(13.46, 114, lower.tail=FALSE)
```

```
## [1] 2.663963e-25
```